*Systematic Review*

# Impact of Diet Interventions for Weight Loss on Lean Body Mass and Resting Metabolic Rate: A Systematic Review

Natallia Lastovka, DCN, RD, CDCES, LDN [1]; Karen L. Stanfar, DCN, MPH, RDN, LD [2]; Mina Ghajar, MLS [3]; Alainn Bailey, DCN, RD [1]; Laura D. Byham-Gray, PhD, RDN, FNKF [1]; Diane R. Radler, PhD, RD [*,1]

[1]School of Health Professions, Rutgers University, Newark, NJ 07101, United States; [2]Ambulatory Nutrition, MetroHealth Medical Center, Cleveland, OH 44109, United States; [3]Health Sciences Libraries, Rutgers University, Newark, NJ 07101, United States

*Corresponding Author: Diane R. Radler, Department of Clinical and Preventive Nutrition Sciences, School of Health Professions, Rutgers University, 65 Bergen Street, Room 171, Newark, NJ 07101-3001, United States (rigassdl@shp.rutgers.edu).

***Context:*** *Individuals with excess weight are at a higher risk of overall health issues and often seek to modify their diets to lose weight. Weight loss can alter body composition, leading to a decrease in lean body mass (LBM) and resting metabolic rate (RMR). Limited research has examined the relationship between different dietary modifications and body weight, LBM, and RMR.*

***Objective:*** *This review aimed to research the impact of weight-loss dietary interventions on body weight, LBM, and RMR in adults with overweight or obesity.*

***Data Sources:*** *The search was conducted using the following databases: CINAHL, Cochrane Library, Embase, PubMed, Scopus, Web of Science, Google Scholar, and Dissertations & Theses Global.*

***Data Extraction:*** *The initial search found 8838 studies. The title and abstract screening identified 138 studies for full-text review. As a result, 16 studies were selected for this systematic review. The studies had to be original, published in English over the last 10 years, and include adults as participants. The intervention had to be any weight-loss dietary modification, and the outcomes had to include measurements of body weight, LBM, and RMR.*

***Data Analysis:*** *The included intervention groups implemented the following dietary modifications: altering calorie intake, macronutrient composition, and meal timing. Across all studies, the dietary interventions resulted in all participants losing a substantial amount of weight. Changes in LBM and RMR, however, were not as consistent across the groups, but there was an overall downward trend. Physical activity did not confound the outcomes of the included studies.*

***Conclusion:*** *Regardless of the type of dietary modification and inclusion of physical activity, any change in diet resulted in significant weight loss. Changes in LBM and RMR varied, with a tendency to decrease. Future studies should research the long-term effects of diet interventions on the outcomes of interest.*



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***Key words:** overweight, obese, weight loss, diet, lean body mass, resting metabolic rate.*



## INTRODUCTION

According to data from global and national statistics, approximately 40% of the population in the world and the United States has excess weight.[1–3] Overweight or obesity occurs when individuals consume more energy than they expend.[4] More than half of those with overweight or obesity pursue dietary interventions to achieve weight loss.[2,5–9]

Weight loss can change the composition of the body, which includes fat mass (FM) and fat-free mass (FFM).[10–16] Fat-free mass includes bone mass, skeletal muscle mass, water, and residual mass.[4,10–12] While the terms FFM and lean body mass (LBM) are often used interchangeably, LBM differs in that it excludes bone tissue, whereas FFM encompasses all nonfat components of the body.[10–12,17] Both terms will be used in this research, depending on how they were reported in the original literature. Negative energy balance impacts lipid storage in muscles and the liver, decreasing FM and LBM.[16,18] Evidence suggests that LBM loss often accompanies FM loss, which can impact overall health and emotional well-being by impairing daily activities, increasing the risk of injury, and reducing energy expenditure (EE).[13–16,19–24] Awareness of the components of body weight and the effects of weight loss on them may help mitigate any risks associated with dieting.

Weight loss can also impact EE. Total EE (TEE) is composed of diet-induced thermogenesis, activity EE, and resting metabolic rate (RMR), which is a major TEE contributor and identifies the energy required for basic metabolic activities.[4,10,16,25,26] Calorimetry measures RMR during the resting state directly or indirectly.[27] Indirect calorimetry is more commonly used because it is noninvasive and estimates RMR by measuring oxygen consumption and carbon dioxide production.[16,28,29] A decrease in RMR will narrow the gap between TEE and caloric intake during restriction, thereby reducing the potential energy deficit required for continued weight loss.

Several factors influence the determination of LBM and RMR.[4,25,30–32] For example, evidence shows that individuals with overweight or obesity expend more energy than individuals of the same stature but with a normal weight, which could be attributed to the body's need for increased FFM to support the additional body weight.[4,25,30–33] Fat-free mass is more metabolically active than adipose tissue, and thus, has a greater impact on RMR: skeletal muscle makes up approximately 23% of RMR, compared with 4% from adipose tissue.[10–12,16,17,34–36] Resting metabolic rate is therefore positively associated with weight and proportional to FFM.[4,26] Males and younger individuals typically have a higher FFM and, consequently, a higher RMR, compared with females and older individuals.[10] Physical activity also supports the maintenance of FFM, which, in turn, sustains RMR.[10,33]

Several studies have shown a decrease in RMR following the initiation of weight loss.[19–22] Heshka et al[37] measured RMR and FFM in adults with obesity and demonstrated that RMR decreased with weight loss due to the loss of FFM. Similarly, research by Leibel et al[32] showed that individuals with obesity who lose 10% of body weight experience a decrease in RMR by 4 kcal per kilogram of FFM. Thus, RMR may be predetermined by body composition, in particular FFM or LBM, and may decrease with weight loss.[16,32,37] Therefore, it is essential to maintain FFM during weight loss to prevent subsequent weight gain during long-term weight management.[10,36,38]

Another mechanism by which weight loss impacts RMR involves the induction of a metabolic adaptation process, which reduces EE beyond what would be expected from the loss of LBM alone.[14,16,19,20,22] This physiological mechanism conserves energy and mainly occurs during a negative energy balance, when measured RMR is lower than what would be estimated based on changes in body composition.[14,16,19,20,22] Energy expenditure declines as an adaptive response to the decrease in energy intake that initiates weight loss as a compensatory reduction in cellular metabolic output, which is understood to conserve LBM.[39,40] Research shows that, during weight loss, EE decreases by approximately 40–120 kcal/day.[19,20] Martins et al[19] found that, after losing 14.1 (±0.4) kg or 13% (±2.8%) of body weight, the metabolic adaptation among participants was −92 (±110) kcal/day ($P < .001$) after 9 weeks in the study, and −38 (±124) kcal/day ($P = .011$) after 13 weeks' duration. Lopez Torres et al[20] demonstrated a reduction of 121 (±188) kcal/day ($P < .001$), after the participants lost 18.4 (±3.9) kg over 10 weeks.

Two contributing factors to decreased EE during weight loss include hormonal changes and malnutrition.[25,41] There may be reduced leptin production in response to a negative caloric balance.[25] Nutritional deficiencies may develop in individuals with excess weight from poor dietary choices and potentially result in malnutrition.[41] Malnourished individuals may exhibit up to a 30% decrease in RMR as a compensatory mechanism.[10,41] Additionally, a subset of individuals with sarcopenic obesity, characterized by low FFM and high FM, might experience further reductions in RMR.[10]

Various dietary modifications might have an impact on body composition.[14] Research has demonstrated a relationship between FFM and energy intake during fluctuations in weight.[16] Nymo et al[42] examined the impact of a very-low-energy diet (2.3 MJ/day for women and 2.8 MJ/day for men) on RMR and body composition. The findings showed that 5% of body-weight loss was associated with significant reductions in FM and RMR.[42] When considering diet and the intake of different macronutrients, the relationship between LBM and RMR becomes relavent as the body metabolizes macronutrients in distinct ways.[43] For example, if energy availability is adequate, protein intake above the recommended dietary allowance may impact the preservation of LBM, specifically the protein synthesis process, including muscle retention and repair.[14,23,44–46]

Previous research has examined various dietary modifications and their impact on body composition and EE. An older meta-analysis of 32 controlled-feeding studies by Hall and Guo[47] examined the effects of modified carbohydrates and fats on body composition and REE, and compared participants with obesity with those of normal weight. The findings showed that lower fat intake increased REE by 26 kcal/day ($P < .0001$) and resulted in greater fat loss (16 g/day; $P < .0001$).[47] Ho et al[48] conducted a systematic review and meta-analysis of 57 studies to study the impact of macronutrient changes on REE during weight loss among adults with overweight or obesity. The findings showed that, during moderate to high weight loss, a 1% increase in carbohydrate intake led to an REE decrease of 2.3 kcal/day (95% CI: −4.11 to −0.47; $P = .013$) and a 1% increase in protein intake led to an REE increase of 3.0 kcal/day (95% CI: 1.02-5.07; $P = .003$).[48]

Other reviews have focused on the relationship between diets and body composition only. Lee and Lee[49] conducted a meta-analysis of 7 studies, exploring the impact of a ketogenic diet combined with exercise among participants with overweight and obesity. The findings demonstrated that the body composition did not show any significant changes.[49] Cheng et al[50] conducted a similar study earlier, specifically a systematic review and meta-analysis of 11 studies, to identify the impact of diet combined with exercise on body composition among peri- and postmenopausal females with overweight or obesity. The dietary interventions included low-calorie modifications (1200–2000 kcal) and modified macronutrient composition of 50%–60% carbohydrate, 10%–20% protein, and 30% fat.[50] The findings showed a significant difference in mean lean mass loss between the diet-only groups and control groups (mean difference: 1.08 (kg); 95% CI: −1.83 to −0.34; $P = .004$).[50] When diet was combined with exercise, LBM loss was greater than with dietary intervention alone (difference in means: −0.84 (kg); 95% CI: −1.13 to −0.55; $P < .001$).[50]

Previous reviews exploring the impact of intermittent fasting on body composition and RMR showed no significant findings. Pureza et al[51] conducted a meta-analysis of 10 studies to study the impact of time-restricted feeding on metabolic parameters, including RMR, among adults with overweight or obesity. The findings showed that there was no significant impact on RMR. [51] Enríquez Guerrero et al[52] conducted a meta-analysis of 18 studies among participants with overweight and obesity studying the impact on intermittent fasting on anthropometrics, body composition, and lipids. The findings showed no significant difference in lean muscle mass loss between the intermittent and continuous diet.[52] Harris et al[53] conducted a systematic review and meta-analysis of 6 studies among patients with overweight and obesity and their impact on anthropometrics and cardiometabolic health, and no significant differences were found in changes in FFM.

Evidence comparing the impact of various weight-loss dietary approaches on body weight, LBM, and RMR is scarce. To address this gap, the main objective of this study was to evaluate recent research on the effects of weight-loss dietary modifications in adults with overweight or obesity on LBM and RMR, providing a comparison of various dietary interventions. This systematic review is valuable because it gathered studies comparing the effects of the most-studied weight-loss interventions on both LBM and RMR. Additionally, it allowed for a side-by-side evaluation and comparison of each intervention on both RMR and LBM. The findings of this review offer a better understanding of changes in body composition and EE, helping to improve weight regulation and the effectiveness of weight-loss strategies for individuals with excess weight. Clinicians and patients might use the insights to help achieve the energy intake needed to promote a healthy weight and ensure weight maintenance once the goal weight is achieved.

## METHODS

### Search Strategy

The research team collaborated with a trained medical librarian (M.G.) to develop the search strategy, using relevant key words (**Appendixes S1** and **S2**). The inclusion criteria for the review were original articles, published in English, and including adults aged 18 years or older as participants.

The initial search was conducted in November 2024, limiting studies to a 10-year time frame (2014–2024), which was selected due to a limited number of relevant studies identified during a preliminary 5-year search and to find articles reflecting the most recent treatments in the management of overweight and obesity. The search was repeated in October 2025 to identify newly published studies that met the inclusion criteria. The search was conducted in PubMed and Scopus, followed by analyzing key words in titles, abstracts, and indexed terms using MeSH (Medical Subject Headings). Once relevant key words were identified, a comprehensive search was performed across multiple databases, including CINAHL (EBSCO), Cochrane Library (Wiley), Dissertations & Theses Global (ProQuest), Embase (Elsevier), PubMed (NLM), Scopus (Elsevier), and Web of Science (Clarivate). Google Scholar was also searched, and the relevant articles were retained. The search results were assessed for relevance, and cited reference searches were performed. Relevant literature reviews were examined for applicable studies, and additional searches were conducted on the primary authors of these studies. All results were organized using EndNote version 20 (Clarivate, London, UK) for citation management.[54]

## Eligibility Criteria

The review included original articles of clinical trials that investigated weight-loss diet interventions with a minimum duration of 1 month. Reviews, observations, and case studies were excluded from the analysis. Table 1 defines the research question and illustrates the criteria used in a study screening process. Each study was analyzed based on the inclusion of a dietary intervention targeting weight loss and 3 measurements: body weight, LBM or FFM, and RMR before and after the intervention. The population included adults with overweight or obesity. If the study intentionally included participants with comorbidities, the study was excluded. For the purpose of this study, the term RMR served as the operational definition of EE at rest. Only studies that explored all 3 outcomes (body weight, LBM/FFM, and RMR) were included. If the study had missing baseline or postintervention data for any of these outcomes, the study was excluded. This approach was to compare differences in each outcome, based on expectations from prior research that RMR changes would depend on LBM loss.

## Literature Search

Titles and abstracts were screened, using EndNote and Rayyan (Rayyan, Cambridge, MA, USA) features.[54,55] A total of 8838 records were identified. After removing duplicates, a title and abstract review was conducted for the remaining 5429 records, during which 5291 studies did not meet the inclusion criteria for the following

**Table 1.** PICOS Criteria for Inclusion and Exclusion of Studies

| Parameter | Inclusion Criteria | Exclusion Criteria |
|---|---|---|
| Participants | • Adults<br>• Overweight or obese<br>• Overall healthy status<br>• Noninstitutional/outpatient setting | • Animal subjects<br>• Pediatric population<br>• Normal weight or underweight<br>  • Note: if the study had >1 group of participants, and 1 of the groups had a normal weight, the study was excluded.<br>• History of specific diseases or chronic conditions<br>• Inpatient population<br>• Specific professions such as military, astronauts, athletes, and others |
| Interventions | • Diet interventions for weight loss | • Bariatric surgery<br>• Hormone-replacement therapy<br>• Weight-loss medication intake<br>• Supplement intake as a diet intervention<br>• Meal replacement as a diet intervention |
| Comparisons | • Comparisons within groups before and after | • Missing baseline or postintervention data |
| Outcomes | • Inclusion of all 3 outcomes:<br>• RMR measurements<br>• LBM or FFM measurements<br>• Body weight<br>• All 3 outcomes to be measured before and after the intervention | • One or more outcomes are missing:<br>• RMR measurements<br>• LBM or FFM measurements<br>• Body weight<br>• Outcomes only measured before the intervention |
| Study design | • Published in English<br>• Clinical trials<br>• ≥ 1 month | • Published in languages other than English<br>• Reviews<br>• Case studies<br>• Observational studies<br>• <1 month |

Abbreviations: FFM, fat-free mass; LBM, lean body mass; RMR, resting metabolic rate.

reasons: nondiet intervention ($n = 2002$), an ineligible population ($n = 1776$), an ineligible study design ($n = 933$), an ineligible outcome ($n = 375$), animal studies ($n = 142$), an ineligible study duration ($n = 48$), and duplicates ($n = 15$). **Figure 1** presents the study selection process in the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flowchart. Full-text screening was conducted in Rayyan by the primary investigator (N.L.) and a second co-investigator (K.L.S.). A third-party reviewer was not required as no conflicts were identified during the screening process. The investigators independently reviewed the full texts of 138 articles and excluded 122 studies due to an ineligible design ($n = 50$), no dietary intervention ($n = 32$), an ineligible outcome ($n = 20$), an ineligible population ($n = 17$), and missing data ($n = 3$). Five articles met the inclusion criteria but had missing data. The team contacted the authors of the articles to request additional data. Of these, only 2 studies provided a complete dataset. As a result, 16 studies met the inclusion criteria and were included in the systematic review (**Appendix S3**).[56–71]

## Quality Control

The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under the registration number CRD420251074910. The research was conducted in accordance with the Cochrane Handbook for Systematic Reviews of Interventions and the PRISMA 2020 Checklist.[72,73] A trained medical librarian (M.G.) supervised the literature search. Two reviewers (N.L., K.L.S.) independently screened titles, abstracts, and full texts, and performed risk-of-bias assessments. The research team met regularly to discuss methods, direction, and results.

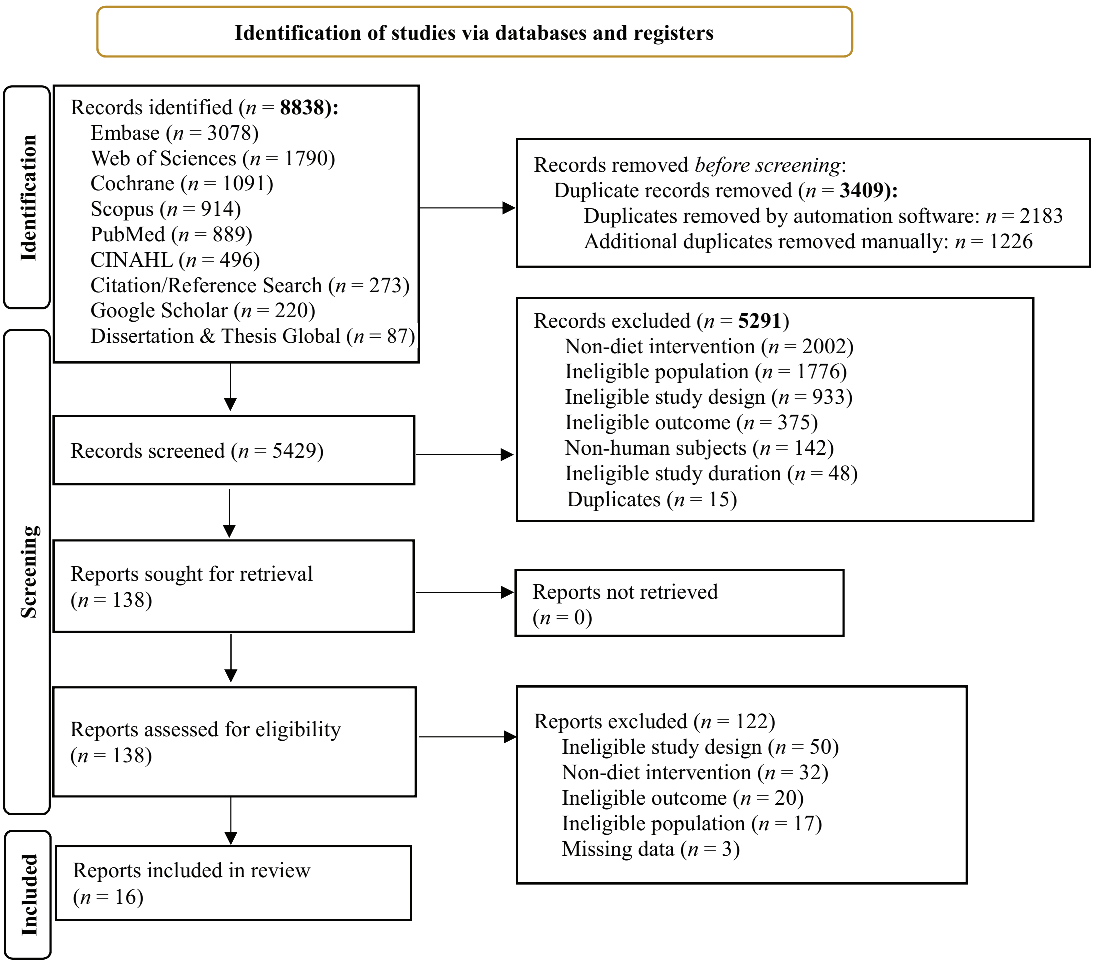


**Figure 1.** PRISMA 2020 Flow Diagram

Abbreviation: PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses. Figure adapted from reference 72.

## Data Extraction and Synthesis

The primary investigator (N.L.) collected data from the selected studies, using an Excel spreadsheet document (Micosoft Corporation, Redmond, WA, USA), and the second reviewer (K.L.S.) reviewed the collected data for accuracy. The extracted data included study characteristics (study citation, authors, year of publication, country, study design, class rating, funding source, study purpose, and sample size), participant characteristics (age, sex, body weight, body mass index [BMI], LBM or FFM, FM, and RMR), intervention characteristics (diet modifications: diet categories such as modified calories, modified meal timing, and modified macronutrients), outcome characteristics (changes in LBM or FFM, RMR, body weight, and measurement tools), confounders (physical activity and duration of the dietary intervention), and findings.

For the narrative synthesis, the reviewers used the Synthesis Without Meta-Analysis (SWIM) reporting guidelines.[74] Five diet categories were initially proposed; however, due to a lack of studies, the 2 *a priori* diet categories, modified food groups and an "other" category, were excluded from the synthesis, and the protocol was revised accordingly. These intervention groups were organized into 3 categories based on the type of diet strategy: modified calories, modified meal timing or intermittent fasting, and modified macronutrient composition, including carbohydrates, proteins, and fats. If the interventions were combined, such as, for example, modified calories and meal timing, they were analyzed in both categories.

The metric chosen for the measured outcomes was kilograms for body weight and LBM or FFM and kilocalories per day for RMR, as the included studies consistently used all of these units. Due to the heterogeneity of outcome measures among the included studies, meta-analysis was not performed. The findings of the included studies were narratively synthesized to address the study objective.

## Risk-of-Bias Assessment

Two reviewers (N.L., K.L.S.) independently assessed the included studies for risk of bias. The Risk of Bias version 2 (RoB 2) tool (Cochrane, London, UK) was used to determine the risk of bias in 13 randomized trials [56–65,67,68,71] (**Figure 2**).[75] The tool uses an outcome-specific approach to assess whether the risk of bias is low, has some concerns, or high.[75] The domains evaluate bias caused by randomization, intervention deviations, incomplete data, outcome assessment, or result selection.[75] The ROBINS-I (Risk Of Bias In Non-randomized Studies of Interventions) tool was used to measure the risk of bias in the remaining 3 nonrandomized studies [66,69,70] (**Figure 3**).[76] ROBINS-I is designed to assess the risk of bias in nonrandomized studies.[76] The tool evaluates bias across confounding and participant selection, intervention classification, intervention deviations, incomplete data, outcome measurement, and results selection.[76] The risk of bias is categorized as low, moderate, serious, or critical risk.[76]

# RESULTS

The included studies enrolled a total of 895 participants who were overweight or obese and exposed them to weight-loss–focused dietary interventions.[56–71] The intervention groups within each study were comparable and did not have significant differences in baseline characteristics.[56–71] The review excluded groups that did not include dietary modifications, such as control groups or groups that included only physical activity as a co-intervention.

## Modified Meal Timing

Out of 16 studies, 4 intervention groups matched the inclusion criteria, specifically looking at all 3 outcomes: body weight, LBM, and RMR, and included modified meal timing into their design, with 3 groups implementing a 16/8-hour intermittent fasting[57,63,67] and 1 group[69] following a 12-hour meal-restricted regimen. Findings showed that meal timing restriction resulted in body-weight reductions ranging from 1.7 to 3 kg or 1.8% to 3.2% (mean body-weight changes; $P < .05$) (**Table 2**, **Appendix S5**). All 4 groups[57,63,67,69] showed nonsignificant reductions in RMR ($P > .05$). Changes in LBM were inconsistent, with half of the groups[57,63] showing a significant decrease in LBM and the remaining half[67,69] reporting a nonsignificant decrease in LBM. None of the groups had an increase in any of the studied outcomes. Physical activity was not included as a co-intervention in any of these studies.[57,63,67,69]

## Modified Calories

**Table 3** and **Appendix S6** list all 24 intervention groups[56–62,64–71] that modified calories to initiate weight loss. Findings show that body weight decreased significantly among all included groups[56–62,64–71] ($P < .05$). Changes in RMR were inconsistent across the groups[56–62,64–71] and no pattern was identified between calorie restriction and changes in RMR. One group[71] showed no change in RMR. Ten[57–59,62,65,69,70] out of 24 groups had nonsignificant changes in RMR, with 7 groups[57–59,62,70] showing a downward trend.

| Study | Risk of Bias Domains* | | | | | |
|---|---|---|---|---|---|---|
| | D1 | D2 | D3 | D4 | D5 | Overall |
| Ashtary-Larky et al. | - | x | + | + | - | x |
| Çelik et al. | - | - | + | + | - | - |
| Cooney et al. | - | - | + | + | - | - |
| Coutinho et al. | - | - | + | + | - | - |
| DeLany et al. | + | - | + | + | - | - |
| Hintze et al. | - | x | + | + | - | x |
| Kleist et al. | - | x | + | + | - | x |
| Lowe et al. | + | - | + | + | + | - |
| McNeil et al. | - | x | + | + | - | x |
| Miller et al. | + | + | + | + | - | - |
| Oldenburg et al. | + | + | + | + | + | + |
| Purcell et al. | - | + | - | + | - | - |
| Yazdanparast et al. | - | - | + | + | - | x |

* Domains: D1: Bias arising from the randomization process; D2: Bias due to deviations from intended interventions; D3: Bias due to missing outcome data; D4: Bias in measurement of the outcome; D5: Bias in selection of the reported result.
Judgement: x high; - some concerns; + low.

**Figure 2.** RoB 2 for Randomized Clinical Trials Included in the Analysis

Abbreviation: RoB 2, Risk of Bias version 2 tool.

| Study | Risk of Bias Domains* | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | D1 | D2 | D3 | D4 | D5 | D6 | D7 | Overall |
| Monteze et al. | - | + | + | + | + | + | + | - |
| Sarica et al. | - | + | + | + | - | + | - | - |
| Stubelj et al. | - | + | + | + | + | - | + | - |

* Domains: D1**:** Bias due to confounding; D2: Bias due to selection of participants; D3: Bias in classification of interventions; D4: Bias due to deviations from intended interventions; D5: Bias due to missing data; D6: Bias in measurement of outcomes; D7: Bias in selection of the reported result.
Judgement: - moderate; + low.

**Figure 3.** ROBINS-I for Non-Randomized Clinical Trials Included in the Analysis

Abbreviation: ROBINS-I, Risk Of Bias In Non-randomized Studies of Interventions.

Almost all groups (23/24)[56–62,64–71] reported a decrease in LBM, with 17 of 23 demonstrating a significant reduction. Similarly, no pattern was observed between the amount of calorie restriction and changes in LBM. Eight out of 24 groups[60,62,64,65,70,71] had physical activity as part of a co-intervention; however, the results did not show any differences between the groups with and without physical activity.

### Modified Macronutrients

*Protein.* The protein dietary modification in the included studies ranged from 10% to 35% of daily calories, as prescribed at enrollment. Most studies reported the actual mean protein intake, which ranged from 16% to 21%. **Table 4** and **Appendix S7** list all groups that had modified protein intake. A significant reduction in body weight was observed across all groups[56,57,59,62,64–66,69–71] ($P < .05$). Change in RMR was inconsistent across the groups[56,57,59,62,64–66,69–71]; however, no pattern was observed in correlation with how much protein was consumed or prescribed. One group[71] showed no change in RMR. Most of the groups (10/19) exhibited nonsignificant changes in RMR,[57,59,62,65,69,70] with 7 groups[56,59,64,66,70] showing a significant decrease ($P < .05$). Only 1 group[65] showed a nonsignificant increase in LBM, while the remaining 18 groups[56,57,59,62,64–66,69–71] reported a decrease in LBM, with 6 of these decreases[57,65,69,71] being

**Table 2.** Effect of Intermittent Fasting on Body Weight, RMR, and LBM

| No. | Study | Year | Duration, wk | *n* | Meal timing restriction or intervention groups | Change in body weight, mean (SD) | Change in RMR, mean (SD) | Change in LBM/FFM, mean (SD) |
|---|---|---|---|---|---|---|---|---|
| 1 | Çelik et al[57] | 2023 | 8 | 10 | 16/8 (fasting/eating hours per day), 10–6 PM, TRE group | −2.3 (1.8) kg, -3.2% (2.6%), $P = .014$* | −53.1 (149.9) kcal, $P = .386$ | −0.8 (0.7) kg, -1.9% (1.8%), $P = .016$* |
| 2 | Lowe et al[63] | 2020 | 12 | 25 | 16/8 (fasting/eating hours per day), 12–8 PM, TRE group, in-person cohort | −1.7 (95% CI: -2.6 to -0.8) kg, -1.8% (-2.9% to -0.8%), $P < .001$* | −28.1 (95% CI: -91.8 to 35.5) kcal, $P = .39$ | −1.1 (95% CI: -1.7 to -0.5) kg, $P < .001$* |
| 3 | Oldenburg et al[67] | 2025 | 12 | 29 | 8-hour eating window, TRE group | −3 (95% CI: -4.1, -1.9) kg, $P < .005$* | −66.4 (95% CI: -154.6 to 21.8) kcal, $P > .05$ | −0.6 (95% CI: -1.5, 0.3) kg, $P > .05$ |
| 4 | Sarica et al[69] | 2022 | 4 | 15 | 12/12 (fasting/eating hours per day), overnight fasting, OFD group | −2.3%, $P = .000$* | –, $P = .893$ | –, $P = .766$ |

*The table lists the studies in alphabetical order of the first author's last name. Significant change ($P < .05$).
Abbreviations: LBM, lean body mass; OFD, overnight fasting diet; RMR, resting metabolic rate; TRE, time-restricted eating.

nonsignificant. Therefore, no pattern was observed between the amount of protein intake and changes in LBM.[56,57,59,62,64–66,69–71] Six out of 19 groups[64,65,70,71] included physical activity as a co-intervention; however, no pattern was observed to suggest that physical activity acted as a confounder in the relationship between protein intake and changes in the studied outcomes.

*Carbohydrates.* The carbohydrate dietary modification in the included studies ranged from 42% to 65% of daily calories, as prescribed at enrollment. Most studies reported the actual mean carbohydrate intake, which ranged from 44% to 54%. **Table 5** and **Appendix S8** list all groups that had modified carbohydrate intake. Body weight significantly decreased across all groups[56,57,59,62,64–66,69–71] ($P < .05$). Changes in RMR were inconsistent: 1 group[71] showed no change, whereas 4 groups[65,69,71] showed an increase, with 1[71] of these increases being statistically significant. Fourteen[56,57,59,64,66,69,70] out of 19 groups showed a decrease in RMR, with half[56,59,64,66,70] of these groups showing a significant decrease. Only 1 group[65] showed a nonsignificant increase in LBM, while the other 18 groups[56,57,59,62,64–66,69–71] showed a decrease in LBM, with 6[57,65,69,71] of these decreases being nonsignificant. Therefore, no pattern emerged between changes in LBM and RMR and how much carbohydrate was consumed or prescribed. Finally, although 6[64,65,70,71] of the 19 groups included physical activity as a co-intervention, there was no evidence that physical activity acted as a confounder in the relationship between carbohydrate intake and changes in the studied outcomes.

*Fat.* The fat dietary modification in the included studies ranged from 20% to 35% of daily calories, as prescribed at enrollment. Most studies reported the actual mean fat intake, which ranged from 26% to 39%. **Table 6** and **Appendix S9** list all groups that had modified fat intake. All groups[56,57,59,62,64–66,69–71] experienced a significant reduction in body weight ($P < .05$). Similar to changes in protein and carbohydrate intakes, variations in RMR and LBM were inconsistent. One group[71] showed no change in RMR, while 4 groups[65,69,71] showed an increase in RMR, with only 1 of these[71] increases being statistically significant. One group[65] demonstrated a nonsignificant increase in LBM, while the other 18 groups[56,57,59,62,64–66,69–71] showed a decrease in LBM. Therefore, no pattern emerged between changes in LBM and RMR and how much fat was consumed or prescribed. Similarly, physical activity was not found to be a confounder in the relationship between fat intake and changes in the studied outcomes.

**Table 3.** Impact of Modified Calories on Body Weight, RMR, and LBM

| No. | Calorie intake, mean (SD) | Intervention group | Study | Year | Duration | $n$ | PA[b] | Change in body weight, mean (SD) | Change in RMR, mean (SD) | Change in LBM/FFM,mean (SD) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1975 (444) ↓ 500[a] | DIET | Purcell et al[68] | 2023 | 12 wk | 20 | –[b] | −3.4 (2.3) kg, $P < .001$* | −48 (21) kcal, $P = .042$* | −1.2 (1.6) kg, $P = .002$* |
| 2 | 1841 (534.3) ↓500[a] | mRMR | Cooney et al[58] | 2021 | 12 wk | 24 | – | −4.2 %, $P < .05$* | $F$ (2.4, 87.2) = 1.5, $P = .220$, partial $\eta^2 = 0.041$ | –, $P < .0005$* |
| 3 | 1645.1 (433.3) ↓500[a] | eRMR | | | | 23 | – | −3.2%, $P < .05$* | | –, $P < .0005$* |
| 4 | 1610.8 (270.3) ↓ 300 kcal[a] | Yoga | Yazdanparast et al[71] | 2020 | 8 wk | 22 | + | −2.2 (1.6) kg, $P < .001$* | 94.6 (104.7) kcal, $P = .001$* | −0.1% (1.3%), $P = .673$ |
| 5 | 1580 (380) | DIW | Kleist et al[62] | 2017 | 12 wk | 44 | + | − 8.8 (4.6) kg, $P < .0001$* | −29 (246) kcal, $P > .05$ | −2.4 (2.2) kg, $P < .0001$* |
| 6 | ↓ 500–800[a] | DI | | | | 38 | – | −7.0% (3.9%), $P < .0001$* | −72 (206) kcal, $P > .05$ | −2.2 (1.8) kg, $P < .0001$* |
| 7 | 1458.7 (108.1) ↓ 500 kcal[a] | Control | Yazdanparast et al[71] | 2020 | 8 wk | 22 | – | −2.8% (1.8%), $P < .001$* | 0.00 (110.2) kcal, $P = 1$ | −0.6 (1.7) kg, $P = .163$ |
| 8 | 1448 (198) | RT+DIET | Miller et al[65] | 2018 | 4 mo | 10 | + | $\beta = -1.7$ (95% CI, −2.5 to −0.9) kg, $P = .0006$* | +, $P = .09$ | $\beta = 0.11$ (95% CI, -0.4, 0.6) kg, $P = .67$ |
| 9 | 1401 (196) | DIET | | | | 10 | – | $\beta =$ -1.4 (95% CI, −2.0 to −0.7) kg, $P = .0004$* | +, $P = .09$ | $\beta = -0.004$, (95% CI, −0.30, 0.29) kg, $P = .98$ |
| 10 | 1255 (58.6) ↓ 33%[a] | CER | Coutinho et al[59] | 2018 | 12 wk | 14 | – | −12.5%, $P < .001$* | –, $P = .193$ | –, $P < .001$* |
| 11 | 1142.64 (300.3) ↓500[a] | ERD | Çelik et al[57] | 2023 | 8 wk | 13 | – | −4.1 (1.4) kg, −5.5 (2.0)%, $P = .001$* | −103.5 (195.0) kcal, $P = .087$ | −0.6 (1.1) kg, -1.3% (2.3%), $P = .087$ |
| 12 | ↓ 500[a] | F | Stubelj et al[70] | 2020 | 6 mo | 20 | + | −4%, $P < .05$* | −7%, $P > .05$ | −2%, $P < .05$* |
| 13 | | M | | | | 13 | + | −4%, $P < .05$* | −10%, $P < .05$* | −2%, $P < .05$* |
| 14 | ↓ 500[a] Minimum 1200[a] | SLOW | Hintze et al[61] | 2019 | 20 wk | 17 | – | −4.5% (4.0%), $P < .001$* | –, $P = .035$* | –, $P < .01$* |
| 15 | ↓ 513–770[a] | One control group | Monteze et al[66] | 2021 | 6 mo | 27 | – | −4.2%, $P < .001$* | −122.9 (29.1) kcal, −8.6 %, $P < .001$* | –, $P < .001$* |
| 16 | ↓500–750[a] | SLOW | Ashtary-Larky et al[56] | 2017 | 15 wk | 18 | – | −5.5 (1.5) kg, $P < .001$* | −22.9 (26.5) kcal, $P < .05$* | −0.5 (0.8) kg, $P < .05$* |
| 17 | ↓ 800[a] | C | McNeil et al[64] | 2015 | 6 mo | 65 | – | −4.8 (4.6) kg, $P < .05$* | −56 (92) kcal, $P < .05$* | −0.9 (2.4) kg, $P < .05$* |
| 18 | Minimum 1000[a] | CR | | | | 28 | + | −6.7 (4.5) kg, $P < .05$* | −67 (110) kcal, $P < .05$* | −0.5 (2.3) kg, $P < .05$* |
| 19 | ↓ 433 (327) | DO | DeLany et al[60] | 2014 | 6 mo | 57 | – | −8.1 (5.9) kg, $P = .0001$* | −114 (140) kcal, $P = .0001$* | −2.1 (2.7) kg, $P = .0001$* |
| 20 | | DPA | | | | 61 | + | −11.0 (7.3) kg, $P = .0001$* | −164 (201) kcal, $P = .0001$* | −2.4 (2.6) kg, $P = .0001$* |
| 21 | ↓ 1000[a] Minimum 1200[a] | RAPID | Hintze et al[61] | 2019 | 10 wk | 19 | – | −6.2% (3.1%), $P < .001$* | –, $P = .035$* | –, $P < .01$* |
| 22 | ↓ 20%[a] | FMD | Sarica et al[69] | 2022 | 4 wk | 15 | – | −2.5%, $P = .003$* | +, $P = .096$ | –, $P = .587$ |
| 23 | ↓ 1000–1500[a] | RAPID | Ashtary-Larky et al[56] | 2017 | 5 wk | 18 | – | −5.1 (1.1) kg, $P < .001$* | −59.3 (32.6) kcal, $P < .001$* | −1.5 (0.8) kg, $P < .001$* |
| 24 | ↓ 15%[a] ↓ 268.7 (95% CI: -576.6, 39.3) | C | Oldenburg et al[67] | 2025 | 12 wk | 29 | – | −3.5 (95% CI, -4.7, -2.3) kg, $P < .05$* | −159.6 (95% CI, -265.3, -53.8) kcal, $P < 0.05$* | −0.6 (95% CI, -1.6, 0.3) kg, $P > .05$ |

*The table lists the studies in descending order of calorie intake. Significant change ($P < .05$).
[a]Prescribed (vs actual).
[b]Denotes if physical activity was included (+) or excluded (-) as part of the intervention.
Abbreviations: C, caloric restriction; CER, continuous energy restriction; CR, caloric restriction and resistance training; DI, diet only; DIET, dietary restriction; DIW, diet and walking; DO, diet alone; DPA, diet plus physical activity; ERD, energy-restricted; eRMR, estimated RMR; F, females; FFM, fat-free mass; FMD, frequent-meal diet; LBM, lean body mass; M, males; mRMR, measured RMR; PA, physical activity; RAPID, rapid-weight-loss group; RMR, resting metabolic rate; RT, resistance training; SLOW, slow-weight-loss group.

**Table 4.** Effect of Protein Intake on Body Weight, RMR, and LBM

| No. | Intervention group | Protein intake, mean % (SD) | Calorie intake, mean (SD) | Study | Year | Duration | *n* | PA[b] | Change in body weight, mean (SD) | Change in RMR, mean (SD) | Change in LBM/FFM, mean (SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | DIW | 21.4 (4.9) g 81.6 (22.2) | 1580 (380) ↓ 500–800[a] | Kleist et al[62] | 2017 | 12 wk | 44 | + | −8.8 (4.6) kg, $P < .0001$* | −29 (246) kcal, $P > .05$ | −2.4 (2.2) kg, $P < .0001$* |
| 2 | DI | 22–25[a] | | | | | 38 | - | −7.0% (3.9%), $P < .0001$* | −72 (206) kcal, $P > .05$ | −2.2 (1.8) kg, $P < .0001$* |
| 3 | One control group | 20–25[a] | ↓ 513–770[a] | Monteze et al[66] | 2021 | 6 mo | 27 | - | −4.2%, $P < .001$* | −122.9 (29.1) kcal, − 8.6%, $P < .001$* | -, $P < .001$* |
| 4 | IER | 19.7 (0.4) 20[a] | 1382 (58.6) ↓ 33%[a] | Coutinho et al[59] | 2018 | 12 wk | 14 | - | −12.5%, $P < .001$* | -, $P < .001$* | -, $P < .001$* |
| 5 | CER | 19.3 (0.4) 20[a] | 1255 (58.6) ↓ 33%[a] | | | | 14 | - | −12.5%, $P < .001$* | -, $P = .193$ | -, $P < .001$* |
| 6 | Yoga | 18.2 (3.5) 15[a] | 1610.8 (270.3) ↓ 300 kcal[a] | Yazdanparast et al[71] | 2020 | 8 wk | 22 | + | −2.2 (1.6) kg, $P < .001$* | 94.6 (104.7) kcal, $P = .001$* | −0.12% (1.3%), $P = .673$ |
| 7 | Control | 17.1 (4.3) 15[a] | 1458.7 (108.1) ↓ 500 kcal[a] | | | | 22 | - | −2.8% (1.8%), $P < .001$* | 0.00 (110.2) kcal, $P = 1$ | −0.57 (1.7) kg, $P = .163$ |
| 8 | ERD | 17.0 (2.5) g 49.7 (19.5) 10–35[a] | 1142.6 (300.3) ↓500[a] | Çelik et al[57] | 2023 | 8 wk | 13 | - | −4.1 (1.4) kg, −5.5% (2.0%), $P = .001$* | −103.5 (195.0) kcal, $P = .087$ | −0.6 (1.1) kg, −1.3% (2.3%), $P = .087$ |
| 9 | TRE | 16.0 (2.3) g 55.2 (10.2) 10–35[a] | 1335.6 (315.9) Unrestricted[a] | | | | 10 | - | −2.3 (1.8) kg, −3.2% (2.6%), $P = .014$* | −53.1 (149.9) kcal, $P = .386$ | −0.8 (0.7) kg, −1.9% (1.8%), $P = .016$* |
| 10 | F | 15–17[a] | ↓ 500[a] | Stubelj et al[70] | 2020 | 6 mo | 20 | + | −4 %, $P < .05$* | −7%, $P > .05$ | −2%, $P < .05$* |
| 11 | M | | | | | | 13 | + | −4%, $P < .05$* | −10%, $P < .05$* | −2%, $P < .05$* |
| 12 | SLOW | 15[a] | ↓500–750[a] | Ashtary-Larky et al[56] | 2017 | 15 wk | 18 | - | −5.5 (1.5) kg, $P < .001$* | −22.9 (26.5) kcal, $P < .05$* | −0.5 (0.8) kg, $P < .05$* |
| 13 | RAPID | | ↓1000–1500[a] | | | 5 wk | 18 | - | −5.1 (1.1) kg, $P < .001$* | −59.3 (32.6) kcal, $P < .001$* | −1.51 (0.80) kg, $P < .001$* |
| 14 | C | 15[a] | ↓ 800[a] Minimum 1000[a] | McNeil et al[64] | 2015 | 6 mo | 65 | - | −4.8 (4.6) kg, $P < .05$* | −56 (92) kcal, $P < .05$* | −0.9 (2.4) kg, $P < .05$* |
| 15 | CR | | | | | | 28 | + | −6.7 (4.5) kg, $P < .05$* | −67 (110) kcal, $P < .05$* | −0.5 (2.3) kg, $P < .05$* |
| 16 | OFD | 10–15[a] | ↓ 20%[a] | Sarica et al[69] | 2022 | 4 wk | 15 | - | −2.3%, $P = .000$* | -, $P = .893$ | -, $P = .766$ |
| 17 | FMD | | | | | | 15 | - | −2.5%, $P = .003$* | +, $P = .096$ | -, $P = .587$ |
| 18 | RT+DIET | 3.1 g/kg of fat-free mass/da | 1448 (198) | Miller et al[65] | 2018 | 4 mo | 10 | + | $\beta = -1.7$ (95% CI, −2.5 to -0.9) kg, $P = .0006$* | +, $P = .09$ | $\beta = 0.11$ (95% CI, -0.40, 0.62) kg, $P = .67$ |
| 19 | DIET | g 112 (29) | 1401 (196) | | | | 10 | - | $\beta = -1.4$ (95% CI, −2.0 to -0.7) kg, $P = .0004$* | +, $P = .09$ | $\beta = -0.004$, (95% CI, -0.30, 0.29) kg, $P = .98$ |

*The table lists the studies in descending order of protein intake. Significant change.
[a]Prescribed (vs actual).
[b]Denotes if physical activity was included (+) or excluded (-) as part of the intervention.
Abbreviations: C, caloric restriction; CER, continuous energy restriction; CR, caloric restriction and resistance training; DI, diet only; DIET, dietary restriction; DIW, diet and walking; ERD, energy-restricted; F, females; FFM, fat-free mass; FMD, frequent-meal diet; IER, intermittent energy restriction; LBM, lean body mass; M, males; OFD, overnight fasting diet; PA, physical activity; RAPID, rapid-weight-loss group; RMR, resting metabolic rate; RT, resistance training; SLOW, slow-weight-loss group; TRE, time-restricted eating.

**Table 5.** Effect of Carbohydrate Intake on Body Weight, RMR, and LBM

| No. | Intervention group | Carbohydrate intake, mean % (SD) | Calorie intake | Study | Year | Duration | *n* | PA[b] | Change in body weight, mean (SD) | Change in RMR, mean (SD) | Change in LBM/FFM, mean (SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | One control group | 55–65[a] | ↓ 513–770[a] | Monteze et al[66] | 2021 | 6 mo | 27 | – | −4.2%, $P<.001$* | −122.9 (29.1) kcal, -8.6 %, $P<.001$* | –, $P<.001$* |
| 2 | OFD | 50–60[a] | ↓ 20%[a] | Sarica et al[69] | 2022 | 4 wk | 15 | – | −2.3%, $P=.000$* | –, $P=.893$ | –, $P=.766$ |
| 3 | FMD | | | | | | 15 | – | −2.5%, $P=.003$* | +, $P=.096$ | –, $P=.587$ |
| 4 | F | >50[a] | ↓ 500[a] | Stubelj et al[70] | 2020 | 6 mo | 20 | + | −4 %, $P<.05$* | −7%, $P>.05$ | −2%, $P<.05$* |
| 5 | M | | | | | | 13 | + | −4%, $P<.05$* | −10%, $P<.05$* | −2%, $P<.05$* |
| 6 | C | 55[a] | ↓ 800[a] Minimum 1000[a] | McNeil et al[64] | 2015 | 6 mo | 65 | – | −4.8 (4.6) kg, $P<.05$* | −56 (92) kcal, $P<.05$* | −0.9 (2.4) kg, $P<.05$* |
| 7 | CR | | | | | | 28 | + | −6.7 (4.5) kg, $P<.05$* | −67 (110) kcal, $P<.05$* | −0.5 (2.3) kg, $P<.05$* |
| 8 | SLOW | 50–55[a] | ↓500–750[a] | Ashtary-Larky et al[56] | 2017 | 15 wk | 18 | – | −5.5 (1.5) kg, $P<.001$* | −22.9 (26.5) kcal, $P<.05$* | −0.5 (0.8) kg, $P<.05$* |
| 9 | RAPID | | ↓1000–1500[a] | | | 5 wk | 18 | – | −5.1 (1.1) kg, $P<.001$* | −59.3 (32.6) kcal, $P<.001$* | −1.51 (0.80) kg, $P<.001$* |
| 10 | Yoga | 54.3 (15.8) 55[a] | 1610.8 (270.3) ↓ 300 kcal[a] | Yazdanparast et al[71] | 2020 | 8 wk | 22 | + | −2.2 (1.6) kg, $P<.001$* | 94.6 (104.7) kcal, $P=.001$* | −0.12% (1.25%), $P=.673$ |
| 11 | CER | 50.6 (0.7) 50[a] | 1255 (58.6) ↓ 33%[a] | Coutinho et al[59] | 2018 | 12 wk | 14 | – | −12.5%, $P<.001$* | –, $P=.193$ | –, $P<.001$* |
| 12 | Control | 49.8 (16.2) 55[a] | 1458.7 (108.1) ↓ 500 kcal[a] | Yazdanparast et al[71] | 2020 | 8 wk | 22 | – | −2.8% (1.8%), $P<.001$* | 0.00 (110.2) kcal, $P=1$ | −0.6 (1.7) kg, $P=.163$ |
| 13 | IER | 48.4 (0.7) 50[a] | 1382 (58.6) ↓ 33%[a] | Coutinho et al[59] | 2018 | 12 wk | 14 | – | −12.5%, $P<.001$* | –, $P<.001$* | –, $P<.001$* |
| 14 | ERD | 47.0 (10.3) g 126.2 (26.5) 45–65[a] | 1142.64 (300.3) ↓500[a] | Çelik et al[57] | 2023 | 8 wk | 13 | – | −4.1 (1.4) kg, −5.5% (2.0%), $P=.001$* | −103.5 (195.0) kcal, $P=.087$ | −0.6 (1.09) kg, -1.3% (2.3%), $P=.087$ |
| 15 | TRE | 45.25 (6.4) g 160.83 (65.0) 45–65[a] | 1335.6 (315.9) Unrestricted[a] | | | | 10 | – | −2.3 (1.8) kg, −3.2% (2.6%), $P=.014$* | −53.1 (149.9) kcal, $P=.386$ | −0.8 (0.7) kg, -1.9% (1.8%), $P=.016$* |
| 16 | DIW | 43.7 (6.8) g 167.1 (44.6) | 1580 (380) ↓ 500–800[a] | Kleist et al[62] | 2017 | 12 wk | 44 | + | −8.8 (4.6) kg, $P<.0001$* | −29 (246) kcal, $P>.05$ | −2.4 (2.2) kg, $P<.0001$* |
| 17 | DI | 42–45[a] | | | | | 38 | – | −7.0% (3.9%), $P<.0001$* | −72 (206) kcal, $P>.05$ | −2.2 (1.8) kg, $P<.0001$* |
| 18 | RT+DIET | g 154 (25) | 1448 (198) | Miller et al[65] | 2018 | 4 mo | 10 | + | $\beta=-1.7$ (95% CI, −2.5 to −0.9) kg, $P=.0006$* | +, $P=.09$ | $\beta=0.1$ (95% CI, −0.4, 0.6) kg, $P=.67$ |
| 19 | DIET | g 143 (25) | 1401 (196) | | | | 10 | – | $\beta=-1.4$ (95% CI, −2.0 to -0.7) kg, $P=.0004$* | +, $P=.09$ | $\beta=$ -0.0, (95% CI, −0.3, 0.3) kg, $P=.98$ |

*The table lists the studies in descending order of carbohydrate intake. *Significant change.
[a]Prescribed (vs actual).
[b]Denotes if physical activity was included (+) or excluded (-) as part of the intervention.
Abbreviations: C, caloric restriction; CER, continuous energy restriction; CR, caloric restriction and resistance training; DI, diet only; DIET, dietary restriction; DIW, diet and walking; ERD, energy-restricted; F, females; FFM, fat-free mass; FMD, frequent-meal diet; IER, intermittent energy restriction; LBM, lean body mass; M, males; OFD, overnight fasting diet; PA, physical activity; RAPID, rapid-weight-loss group; RMR, resting metabolic rate; RT, resistance training; SLOW, slow-weight-loss group; TRE, time-restricted eating.

**Table 6.** Effect of Fat Intake on Body Weight, RMR, and LBM

| No. | Intervention group | Fat intake, mean % (SD) | Calorie intake, mean (SD) | Study | Year | Duration | *n* | PA[b] | Change in body weight, mean (SD) | Change in RMR, Mean (SD) | Change in LBM/FFM, mean (SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | TRE | 38.5 (4.3) g 57.7 (15.8) 20–35[a] | 1335.6 (315.9) Unrestricted[a] | Çelik et al[57] | 2023 | 8 wk | 10 | – | −2.3 (1.8) kg, −3.2% (2.6%), $P = .014$* | −53.1 (149.9) kcal, $P = .386$ | −0.8 (0.7) kg, −1.9% (1.8%), $P = .016$* |
| 2 | ERD | 35.0 (10.3) g 46.0 (13.9) 20–35[a] | 1142.64 (300.3) ↓500[a] | | | | 13 | – | −4.1 (1.4) kg, −5.5% (2.0%), $P = .001$* | −103.5 (195.0) kcal, $P =$ . $P = .087$ | −0.6 (1.09) kg, −1.3% (2.3%), $P = .087$ |
| 3 | SLOW | 30–35[a] | ↓500–750[a] | Ashtary-Larky et al[56] | 2017 | 15 wk | 18 | – | −5.5 (1.5) kg, $P < .001$* | −22.9 (26.5) kcal, $P < .05$* | −0.5 (0.8) kg, $P < .05$* |
| 4 | RAPID | | ↓1000–1500[a] | | | 5 wk | 18 | – | −5.1 (1.1) kg, $P < .001$* | −59.3 (32.6) kcal, $P < .001$* | −1.51 (0.80) kg, $P < .001$* |
| 5 | DIW | 32.7 (6.4) g 56.0 (19.9) 32–35[a] | 1580 (380) ↓ 500–800[a] | Kleist et al[62] | 2017 | 12 wk | 44 | + | − 8.8 (4.6) kg, $P < .0001$* | −29 (246) kcal, $P > .05$ | −2.4 (2.2) kg, $P < .0001$* |
| 6 | DI | | | | | | 38 | – | −7.0% (3.9%), $P < .0001$* | −72 (206) kcal, $P > .05$ | −2.2 (1.8) kg, $P < .0001$* |
| 7 | C | 30[+] | ↓ 800[a] Minimum 1000[a] | McNeil et al[64] | 2015 | 6 mo | 65 | – | −4.8 (4.6) kg, $P < .05$* | −56 (92) kcal, $P < .05$* | −0.9 (2.4) kg, $P < .05$* |
| 8 | CR | | | | | | 28 | + | −6.7 (4.5) kg, $P < .05$* | −67 (110) kcal, $P < .05$* | −0.5 (2.3) kg, $P < .05$* |
| 9 | One control group | 25–30[a] | ↓ 513–770[a] | Monteze et al[66] | 2021 | 6 mo | 27 | – | −4.2%, $P < .001$* | −122.9 (29.1) kcal, −8.6%, $P < .001$* | –, $P < .001$* |
| 10 | OFD | 25–30[a] | ↓ 20%[a] | Sarica et al[69] | 2022 | 4 wk | 15 | – | −2.3%, $P = .000$* | –, $P = .893$ | –, $P = .766$ |
| 11 | FMD | | | | | | 15 | – | −2.5%, $P = .003$* | +, $P = .096$ | –, $P = .587$ |
| 12 | F | 25–30[a] | ↓ 500[a] | Stubelj et al[70] | 2020 | 6 mo | 20 | + | −4%, $P < .05$* | −7%, $P > .05$ | −2%, $P < .05$* |
| 13 | M | | | | | | 13 | + | −4%, $P < .05$* | −10%, $P < .05$* | −2%, $P < .05$* |
| 14 | IER | 27.5 (0.7) 30[a] | 1382 (58.6) ↓ 33%[a] | Coutinho et al[59] | 2018 | 12 wk | 14 | – | −12.5%, $P < .001$* | –, $P < .001$* | –, $P < .001$* |
| 15 | Yoga | 27.1 (6.9) 30[a] | 1610.8 (270.3) ↓ 300 kcal[a] | Yazdanparast et al[71] | 2020 | 8 wk | 22 | + | −2.2 (1.6) kg, $P < .001$* | 94.6 (104.7) kcal, $P = .001$* | −0.12% (1.25%), $P = .673$ |
| 16 | Control | 27.2 (6.6) 30[a] | 1458.7 (108.1) ↓ 500 kcal[a] | | | | 22 | – | −2.8% (1.8%), $P < .001$* | 0.0 (110.2) kcal, $P = 1$ | −0.6 (1.7) kg, $P = .163$ |
| 17 | CER | 26.1 (0.7) 30[a] | 1255 (58.6) ↓ 33%[a] | Coutinho et al[59] | 2018 | 12 wk | 14 | – | −12.5%, $P < .001$* | –, $P = .193$ | –, $P < .001$* |
| 18 | RT+DIET | 20[a] g 41 (6) | 1448 (198) | Miller et al[65] | 2018 | 4 mo | 10 | + | $\beta =$ -1.7 (95% CI, −2.5 to -0.9) kg, $P = .0006$* | +, $P = .09$ | $\beta = 0.1$ (95% CI, −0.4, 0.6) kg, $P = .67$ |
| 19 | DIET | 20[a] g 42 (5) | 1401 (196) | | | | 10 | – | $\beta =$ -1.4 (95% CI, −2.0 to -0.7) kg, $P = .0004$* | +, $P = .09$ | $\beta =$ -0.0, (95% CI, −0.3, 0.3) kg, $P = .98$ |

The table lists the studies in descending order of fat intake. *Significant change.
[a]Prescribed (vs actual).
[b]Denotes if physical activity was included (+) or excluded (-) as part of the intervention.
Abbreviations: C, caloric restriction; CER, continuous energy restriction; CR, caloric restriction and resistance training; DI, diet only; DIET, dietary restriction; DIW, diet and walking; ERD, energy-restricted; F, females; FFM, fat-free mass; FMD, frequent-meal diet; IER, intermittent energy restriction; LBM, lean body mass; M, males; OFD, overnight fasting diet; PA, physical activity; RAPID, rapid-weight-loss group; RMR, resting metabolic rate; RT, resistance training; SLOW, slow-weight-loss group; TRE, time-restricted eating.

*Risk-of-Bias Assessment.* The risk of bias for 13 randomized trials,[56–65,67,68,71] assessed using the RoB 2 tool, identified 5 of the included studies[56,61,62,64,71] as having a high level of bias due to deviations from the intended interventions (**Figure 2**). Seven studies[57–60,63,65,68] were rated as "some concerns" for potential bias, and only 1 study[67] had a low overall risk of bias. The remaining 3 nonrandomized studies,[66,69,70] assessed with the ROBINS-I tool, were found to have a moderate overall risk of bias due to confounding factors, missing data, outcome measurement, and selection of the reported results (**Figure 3**).

## DISCUSSION

This study researched the impact of weight-loss dietary interventions on body weight, LBM, and RMR among adults with overweight or obesity, which, to our knowledge, has not been conducted in prior research, making this study notable. The findings demonstrated that all dietary modifications, including modified calorie intake, modified meal timing, and modified macronutrients, or a combination of these, resulted in a significant reduction in body weight. The initial expectation was to find a clear dependency between these interventions and changes in LBM and RMR outcomes; however, such a dependency was not observed. Changes in RMR and LBM were not as consistent, even though most groups demonstrated an overall downward trend in these outcomes.

All dietary interventions led to weight loss, with body-weight reductions ranging from 2 to 11 kg. The daily calorie intake ranged from 1100 to 1900 calories across the intervention groups (**Appendix S5**), but the degree of caloric restriction was not associated with the studied outcomes. The reductions in LBM and FFM ranged from 1 to 2 kg, and the decrease in RMR was between 29 and 160 calories. This questions whether these results are meaningful and clinically significant.

According to current evidence, modified calorie intake and modified macronutrient composition may promote weight loss, whereas the modified timing of eating requires further research in the literature.[6,77] Calorie restriction remains the most popular dietary intervention for weight loss, and data show that reducing daily calories by one-third or by 500 calories for females and 750 calories for males promotes at least a 3% body-weight loss.[6,14,77]

In this study, no association was found between the amounts of carbohydrates, proteins, and fats and the outcomes being studied. Carbohydrate intake ranged from 43% to 54% of total daily calories among the participants in the included studies, and some intervention groups consumed between 126 and 167 g. However, some intervention groups were prescribed up to 65% of total daily energy (**Appendix S8**). Previous research has found mixed results. For example, a review of clinical trials by Smith et al[78] reported that the quantity of carbohydrates did not have a significant impact on energy balance and weight-loss outcomes, which aligns with the findings of this review. In contrast, other studies have suggested that the most effective strategies involve limiting daily carbohydrate intake to 20–150 g or to 40% or less of total calories, alongside increasing protein to 30% of total daily energy.[6,14,77,79] A recent systematic review and meta-analysis by Ho et al[48] found that, during moderate or rapid weight loss, every 1% increase in carbohydrate consumption may lead to a daily reduction in RMR by 2.3 calories, while every 1% increase in protein and fat consumption may increase RMR by 3 calories. Additional literature suggests that weight loss is more successful when limiting foods high in fat or carbohydrates, or those low in fiber.[77]

Prior studies defined a low-fat diet as providing approximately 20% of total calories from fat, and a lower-fat diet was described as containing less than 30% fat.[14,77] A meta-analysis of 32 controlled-feeding studies demonstrated that EE and fat loss were more favorable with lower-fat dietary manipulations.[47] In addition, this correlation might be gender-dependent: LBM was positively associated with dietary fat and protein intake in females only.[80] In the current study, the fat intake ranged from 20% to 35% (prescribed) and 26%–38% (actual) of total calories (**Appendix S9**). Prescribed protein ranged from 10% to 25% of total energy, and the actual intake range was from 16% to 21% (**Appendix S7**). Previous findings have reported that higher consumption of dietary protein may have an LBM-preserving effect by activating muscle protein synthesis and increasing satiety during a calorie deficit.[14,44–46,81] Additionally, increased intake of carbohydrates and protein may be associated with an increased thermic effect of food, which impacts EE and, thus, weight control.[81,82] One review examined the current evidence on diets that have modified macronutrients, such as high-protein and low-carbohydrate diets, as well as meal timing, including intermittent fasting, among individuals seeking weight loss.[2] The author concluded that different dietary approaches led to similar weight loss outcomes, and no specific diet was found to be more effective.[2]

In summary, the study's findings demonstrate a wide variability in macronutrient intake across the intervention groups; however, no pattern was defined in the current study. The short duration of the included studies was partially the reason, since some of the studies were not long enough to show a significant change in the outcomes. No relationship was found between the meal timing and the outcomes. Previous studies have shown an association with the circadian rhythm.[83] However, the overall impact is limited and may be similar to the effects of modified calories.[6,14] In this study, no definitive conclusions could be drawn due to the small number of intervention groups.

### Limitations

The included studies differed in their duration, intervention, sample size, participants' gender, and age. Data extraction included actual and prescribed calories and macronutrients, as not all studies provided the actual intake of these nutrients. Additionally, the studied outcomes were often reported with raw data, as the percentage of changes was not provided, which raises questions about the clinical significance of the observed changes. High attrition rates may raise a concern regarding adherence to the interventions. A moderate to high risk of bias demonstrated by almost all included studies may be a limitation in the quality of the study results.

Future studies should include large samples and investigate potential confounders, such as cultural differences, age, sex, medication intake, and other relevant factors.[5,84–86] The association between physical activity and LBM and RMR needs to be further investigated. Only 5 of the 16 studies included physical activity as part of their interventions, and this current review did not distinguish between different types of physical activity, which may be related to the reason why physical activity did not confound the studied outcomes.

### Implications of Findings

All 3 categories of included dietary modifications resulted in significant body-weight loss, but no relationship was found between any specific diet and changes in LBM and RMR. Individuals with overweight or obesity and health providers who treat patients with excess weight could use these findings to choose any of these interventions for weight loss as long as it is safe. Tailoring the intervention to individuals' preferences will increase adherence and contribute to long-term weight maintenance.

## CONCLUSION

All intervention groups experienced a significant decrease in body weight, regardless of the type of dietary modification or inclusion of physical activity. Changes in RMR and LBM were inconsistent, but most groups demonstrated a downward trend. No pattern emerged between specific dietary interventions and changes in RMR and LBM. Physical activity was not a confounder in the review. Future research should focus on the measurement of body composition as opposed to only body weight and identify strategies to prevent the loss of LBM and RMR. Additionally, the long-term effects of dietary interventions on body composition and RMR should be investigated.


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### Supplementary Material

Supplementary Material is available at *Nutrition Reviews* online.


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### Conflicts of Interest

None declared.

### Data Availability

This study does not have original data available.

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